Nominal Performance

From engineering requirements to weighted H∞ bounds

"Track slowly varying references, reject load disturbances, ignore sensor noise, and do not overwork the actuator." None of these sentences specifies a controller. Each must first become a statement about **which inputs are allowed** and **which outputs are acceptable**. Weighting is the translation, not an extra tuning trick.

> The construction is
>
> physical signals ⟶ normalized budgets ⟶ weighted closed-loop map ⟶ generalized plant.
>
> Internal stability remains a separate requirement. A small weighted transfer norm does not certify hidden internal modes, step overshoot, or actuator saturation.

1 One norm, two different experiments

Let G be a real-rational, proper, stable transfer matrix, realized with decaying internal transients. Use Euclidean vector norms and their induced matrix norms. With zero initial state, its finite-energy gain is

$$\|G\|_\infty = \sup_{w\in L_2, w\neq 0} \frac{\|Gw\|_2}{\|w\|_2} = \sup_{\omega\in\mathbb{R}} \sigma_{\max(G(i\omega))}. \tag{1}$$

The first experiment measures total energy. A persistent sinusoid has infinite energy, so substituting it in this ratio is invalid. Yet the **same number** governs its steady-state RMS gain (book Theorems 5.1–5.2).

For the complex exponential $w(t) = ae^{i\omega t}$, stability gives

$$z_{s(t)} = G(i\omega)ae^{i\omega t}, \quad \frac{\|z_s\|_{\mathrm{RMS}}}{\|w\|_{\mathrm{RMS}}} = \frac{\|G(i\omega)a\|}{\|a\|}. \tag{2}$$

Here RMS means the square root of the long-time average of w^*w; it equals $\|a\|$ for this complex signal. Maximizing over a selects the largest singular value at that frequency; maximizing over frequency gives $\|G\|_\infty$.

This is a steady-state statement, not a bound on the maximum instantaneous output. An initial transient disappears from the long-time average but may matter in a real tracking maneuver. The energy experiment and the sinusoidal experiment agree on their worst gain, not on their admissible signals or their transient behavior.

2 Which sinusoid is worst? Include the endpoints

At a fixed nonzero frequency, choose a unit **right** singular vector v such that

$$G(i\omega)v = \sigma_{\max(G(i\omega))}q, \quad \|q\| = \|v\| = 1. \tag{3}$$

These vectors are generally complex. The physical input is $w(t) = \Re(ve^{i\omega t})$: its components can have different phases. Its real RMS is $\frac{\|v\|}{\sqrt{2}}$, and the output RMS is $\frac{\|G(i\omega)v\|}{\sqrt{2}}$, so the factor cancels in the gain. Restricting v to real vectors at nonzero frequency incorrectly forbids relative phase shifts. Equivalently, multiply the real sinusoid by $\sqrt{2}$ to obtain unit RMS.

For example, the stable row $G(s) = \left[1, \frac{1-s}{1+s}\right]$ has at $\omega = 1$ the value $[1, -i]$. Its complex unit direction $v = \frac{(1,i)^T}{\sqrt{2}}$ gives gain $\sqrt{2}$. Every real unit vector gives only gain 1 at this frequency. The missing gain comes from coordinating the two input phases, not increasing either input budget.

At zero frequency, $G(0)$ is real and a real maximizing singular vector exists. The input is a constant (a step applied from $t = 0$), with RMS $\|a\|$, **not** $\frac{\|a\|}{\sqrt{2}}$. The zero-frequency gain is the final-value ratio, not the step-response peak. For $G(s) = \frac{1}{s+1}$ the gain 1 occurs at zero.

At infinite frequency, properness gives $G(i\omega) \to D$. Infinity is a limit, not a realizable sinusoid. For $G(s) = \frac{s}{s+1}$,

$$|G(i\omega)| = \frac{\omega}{\sqrt{1+\omega^2}} < 1 \tag{4}$$

for every finite positive ω, but $\|G\|_\infty = 1$. Thus use a **supremum**, not a guaranteed finite-frequency maximum. Frequencies approaching infinity approach the bound. A dense finite frequency grid alone does not prove an H_∞ bound.

2.1 Why adding frequencies cannot make the gain worse

For pairwise distinct frequencies, let $w = \sum_j a_j e^{i\omega_j t}$. Time-averaged cross terms vanish, giving

$$\|w\|_{\mathrm{RMS}}^2 = \sum_j \|a_j\|^2, \quad \|z_s\|_{\mathrm{RMS}}^2 = \sum_j \|G(i\omega_j)a_j\|^2 \le \|G\|_\infty^2 \sum_j \|a_j\|^2. \tag{5}$$

One sinusoid already attains or approaches the supremum. For real sums, use DC plus positive-frequency real sinusoids; each nonzero-frequency term contributes $\frac{\|a_j\|^2}{2}$. Do not count a conjugate pair twice under that convention.

> A worst-case disturbance chooses both a frequency and a coordinated spatial/phase direction. Looking only at individual transfer entries can miss it.

3 Weights have two different jobs

Write the physical closed-loop map as $z = Fw$, where $F = S(P, K)$ in the book's notation. Do not confuse this interconnection symbol with sensitivity S below. Choose stable proper real-rational filters and define

$$w = W_w \tilde{w}, \quad \tilde{z} = W_z z, \quad \tilde{z} = W_z F W_w \tilde{w}. \tag{6}$$

Then, for the internally stable closed loop,

$$\|\tilde{w}\|_{\mathrm{RMS}} \le 1 \Rightarrow \|\tilde{z}_s\|_{\mathrm{RMS}} \le 1 \quad \leftrightarrow \quad \|W_z F W_w\|_\infty \le 1. \tag{7}$$

The equivalence is for all finite sinusoidal sums, or alternatively for all finite-energy inputs with the corresponding energy norm.

Input shaper W_w	Large magnitude means **more physical input is allowed** for a unit normalized input. A low-pass shaper models predominantly slow disturbances.
Output weight W_z	Large magnitude means **less physical output is tolerated**. It measures output relative to an error allowance.

For scalar weights, at a frequency where their product is nonzero,

$$|F(i\omega)| \le \frac{1}{|W_{z(i\omega)} W_{w(i\omega)}|}. \tag{8}$$

Thus the tolerated envelope is the **inverse** of the output weight, not the weight itself. A constant error tolerance ε requires $W_z = \frac{1}{\varepsilon}$. A desired envelope need not have a stable proper rational inverse; practical weights approximate it while preserving a valid realization.

3.1 A factor-of-ten error in interpretation

Suppose $F = 0.2$ and the disturbance amplitude allowance is $A = 0.1$, while the output tolerance is $E = 0.05$. Choose $W_w = A$, $W_z = \frac{1}{E} = 20$:

$$\|W_z F W_w\| = 20 \times 0.2 \times 0.1 = 0.4 \le 1. \tag{9}$$

The largest physical output is 0.02, safely below 0.05. If instead $A = 1$, the same tolerance requires $|F| \le 0.05$; the actual map fails with weighted gain 4. Using $W_z = E$ would report the misleading value 0.01. Worse, replacing the small input shaper $A = 0.1$ by its inverse would report 40 and reject the feasible first requirement. These are different physical problems.

For matrix weights, keep the order: at each frequency the input set is

$$\{W_w a : \|a\| \le 1\} \tag{10}$$

and the acceptable output set is

$$\{z : \|W_z z\| \le 1\}. \tag{11}$$

These are complex amplitude ellipsoids (possibly degenerate or unbounded). No invertibility assumption is needed for the weighted map itself; a zero or singular weight simply leaves some directions untested.

4 Tracking, disturbance rejection, and actuator effort

To make every sign explicit, use a SISO negative-feedback loop

$$y_p = Gu + d, \quad y_m = y_p + n, \quad u = C(r - y_m), \quad e = r - y_p. \tag{12}$$

Here d is an **output** disturbance and n is sensor noise. Define

$$S = \frac{1}{1+GC}, \quad T = G\frac{C}{1+GC}, \quad S + T = 1. \tag{13}$$

Eliminating the loop gives

$$e = Sr - Sd + Tn, \quad y_p = Tr + Sd - Tn, \quad u = CS(r - d - n). \tag{14}$$

If d entered at the plant input, its output channel would instead be GS. If error meant $r - y_m$, its noise channel would be $-S$, not T. Signal location and the definition of "error" are part of the specification.

Slow tracking and output-load rejection ask for small S at low frequency. Sensor-noise rejection asks for small T at high frequency. Low-frequency S small also makes T close to 1: this is how an **upper** bound on one channel enforces approximate transmission in its complement. It is not an independent lower singular-value constraint imposed by an upper norm bound.

With $w = (d, n, r)^T$, $W_w = \mathrm{diag}(W_d, W_n, W_r)$, one useful normalized map is

$$M = \begin{pmatrix} -W_e S W_d & W_e T W_n & W_e S W_r \\ -W_u C S W_d & -W_u C S W_n & W_u C S W_r \end{pmatrix}. \tag{15}$$

The first output row penalizes error, the second actuator command. Constant $W_u = \frac{1}{u_{\text{allow}}}$ puts effort in units of its allowance. A dynamic W_u can penalize fast actuator activity more strongly. The bound $\|M\|_\infty \le 1$ is an RMS/energy effort budget, **not** the pointwise saturation constraint $|u(t)| \le u_{\max}$ for arbitrary inputs.

4.1 A concrete loop, not a synthesis claim

Take $G = \frac{1}{s+1}$ and $C = 9$. Then

$$S = \frac{s+1}{s+10}, \quad T = \frac{9}{s+10}, \quad CS = 9\frac{s+1}{s+10}. \tag{16}$$

Use $W_S = \frac{2}{s+0.2}$, $W_T = \frac{s+1}{2(s+10)}$, and $W_u = 0.1$. Here $W_S S$ permits $|S(0)| \le 0.1$ and relaxes with frequency; W_T penalizes high-frequency T more than low-frequency T. The familiar mixed-sensitivity stack $[W_S S; W_T T; W_u CS]$ has gain about 1.00505: it **fails** a unit joint bound even though each channel individually passes. The next page explains why. These weights are an illustrative test, not an assertion that this controller optimizes them.

5 A joint budget is not several independent budgets

For normalized inputs w_1, w_2, w_3, the unit ball

$$\sum_j \|w_j\|_{\mathrm{RMS}}^2 \le 1 \tag{17}$$

is a **smaller** uncertainty set than three separate unit bounds. It implies each individual norm is at most 1, but the converse fails. Three separate squared bounds $\le \frac{1}{3}$ imply the joint bound, not conversely. These inclusions concern **input sets**: protecting the bigger set is the stronger performance requirement.

Consider the static row $z = 0.6w_1 + 0.6w_2$. Its gain for a stacked unit input is $\sqrt{0.6^2 + 0.6^2} = 0.848528$, so a unit output tolerance passes. With **independent** unit inputs, coherent $w_1 = w_2 = 1$ gives $z = 1.2$: it fails. Testing each channel alone gives 0.6 and misses their cooperation.

The same issue appears when **outputs** are stacked. For a scalar input and two outputs $z_1 = z_2 = 0.8w$, each separate gain is 0.8, but the stack gain is $\sqrt{2} \times 0.8 = 1.131371$. In general, for m blocks,

$$\max_j \|M_j\|_\infty \le \left\| \begin{pmatrix} M_1 \\ \dots \\ M_m \end{pmatrix} \right\|_\infty \le \sqrt{\sum_j \|M_j\|_\infty^2}. \tag{18}$$

The upper bound need not be attained: different channels can peak at different frequencies or input directions. A unit stack bound guarantees every individual bound, but is usually more demanding. Allocate budgets deliberately rather than silently replacing separate requirements with a stack.

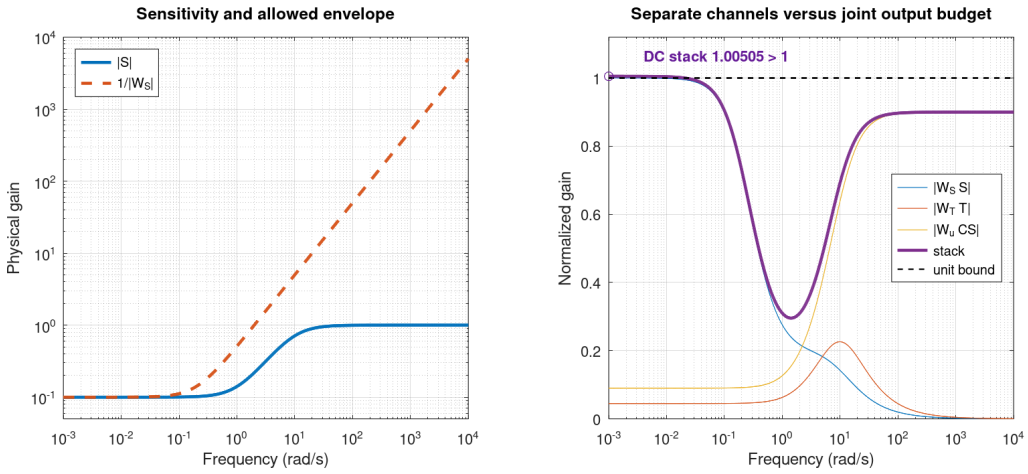


Figure 1: Actual computations for the loop on the previous page. Left: physical sensitivity and its inverse-weight ceiling. Right: every weighted channel stays at or below 1, but their Euclidean stack exceeds 1 near DC. Endpoint calculations, not the grid alone, certify this.

At DC, the squared stack gain is exactly

$$1^2 + 0.045^2 + 0.09^2 = 1.010125. \tag{19}$$

The script proves its squared magnitude never exceeds the DC value using a nonnegative polynomial in ω^2. Its exact norm is $\sqrt{1.010125} = 1.005050$, attained at zero, while the effort channel approaches 0.9 at infinity.

6 Matrix weights are geometry, not scalar ceilings

A scalar weight commutes with a scalar transfer. Matrix weights generally do not commute, and bounding products of their norms is only a sufficient test. At one frequency (or for these constant stable transfers), take

$$F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad W_z = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad W_w = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}. \tag{20}$$

The correct map is

$$W_z F W_w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \|W_z F W_w\| = 1. \tag{21}$$

But moving the input weight through F gives

$$W_z W_w F = \begin{pmatrix} 0 & 2 \\ \frac{1}{2} & 0 \end{pmatrix}, \quad \|W_z W_w F\| = 2. \tag{22}$$

The first direction is allowed amplitude 1 and lands in the output direction with tolerance 1. The second is allowed amplitude $\frac{1}{2}$ and lands in the output direction with tolerance $\frac{1}{2}$. Both are exactly acceptable. Reordering loses this directional correspondence and declares a feasible requirement infeasible. Likewise, $\|W_z\|\|F\|\|W_w\| = 2$ is a conservative bound, not the answer.

6.1 Matching a desired response includes phase

Sometimes "small output" is the wrong objective: the output should resemble a specified stable proper model W_m. Require

$$\|W_1(F - W_m)W_2\|_\infty \le \gamma, \quad \gamma > 0. \tag{23}$$

Replace W_1 by $\frac{W_1}{\gamma}$ to normalize the threshold to 1. Subtract the model **before** taking magnitudes: $|F| - |W_m|$ is not $|F - W_m|$. For instance, $F = -1$ and $W_m = 1$ have identical Bode magnitudes but discrepancy 2, so a tolerance of 0.1 fails by a factor of 20. Model matching controls complex frequency-response error, hence amplitude **and** phase indirectly.

A dynamic exact check: if $F = \frac{1}{s+1}$ and $W_m = \frac{2}{s+2}$, then

$$F - W_m = -\frac{s}{(s+1)(s+2)}, \quad |F(i\omega) - W_{m(i\omega)}|^2 = \frac{\omega^2}{(\omega^2+1)(\omega^2+4)}. \tag{24}$$

Putting $x = \omega^2$, differentiation gives a numerator $4 - x^2$, so the peak occurs at $\omega = \sqrt{2}$ with gain $\frac{1}{3}$. An unweighted tolerance 0.4 passes; 0.3 fails. Here both endpoints have zero discrepancy: looking only at DC and high-frequency limits would miss the violation.

> Weights normalize tolerances; they do not guarantee feasibility. Check both frequency and direction, and examine the physical unweighted responses after design. A norm minimizer need not "hug" the entire envelope.

7 Build the generalized plant without changing the feedback

Use the book's positive interconnection $u = Ky$ (the corrective sign is inside K). Partition the physical open loop as

$$\begin{pmatrix} z \\ y \end{pmatrix} = \begin{pmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{pmatrix} \begin{pmatrix} w \\ u \end{pmatrix}, \quad F = P_{11} + P_{12}K(I - P_{22}K)^{-1}P_{21}. \tag{25}$$

With $w = W_2\tilde{w}$ and performance output $\tilde{z} = W_1(z - W_m w)$, direct substitution gives

$$\tilde{P} = \begin{pmatrix} W_1(P_{11} - W_m)W_2 & W_1P_{12} \\ P_{21}W_2 & P_{22} \end{pmatrix}. \tag{26}$$

Closing the same controller yields

$$S(\tilde{P}, K) = W_1(S(P, K) - W_m)W_2. \tag{27}$$

Thus model matching and weighted gain bounds ($W_m = 0$) share the same synthesis format. No inverse weights need be implemented. The reference-model branch is driven by the same physical exogenous input w, not by controller output.

7.1 When does this preserve the stabilizer class?

The unchanged P_{22} preserves the controller's measurement-to-actuation feedback equations and its algebraic well-posedness condition. This is **not**, by itself, a theorem about arbitrary state-space realizations. Internal stability also tests autonomous states that can be invisible in every displayed transfer matrix.

A safe implementation retains the original plant states and appends **Hurwitz** realizations of the proper stable input shapers, output filters, and model. No appended state feeds back from a weighted performance output to y or u. After eliminating a well-posed algebraic loop and ordering states as input shapers $\longrightarrow$ physical closed loop $\longrightarrow$ model/output filters, the state matrix is block triangular with the original closed-loop matrix and Hurwitz diagonal blocks. The model is a stable branch driven by the shaper; it may be placed before the output filter. Therefore the augmented matrix is Hurwitz **if and only if** the original closed-loop matrix is Hurwitz. Stable unobservable appended modes are harmless; cancellations of unstable hidden states are not.

In the book's language, Hypothesis 5.3 requires both interconnections to be generalized plants; Lemma 5.4 then invokes the Chapter 3 internal-stability test to identify their stabilizing controllers via the common P_{22}. Check that hypothesis, or use the explicit stable cascade argument above. Do not extend the lemma to arbitrary nonminimal realizations merely because their transfers have the same lower-right block.

For a counterexample, append an unconnected state $\dot{q} = q$ to an otherwise stable weighted plant. All transfer blocks, including P_{22}, are unchanged, yet no controller stabilizes that state. Likewise, implementing a "stable" weight by unstable pole-zero cancellation is not a Hurwitz realization. An ideal integrator weight is not stable either: use a leaky approximation or analyze the augmented stabilization problem explicitly rather than quoting equivalence.

8 Three worked self-checks and a reproducible workflow

Chapter 5 contains no book exercises. These are original checks with solutions, chosen to distinguish interpretations that can lead to different design decisions.

8.1 Is the specification impossible before synthesis starts?

Ask for $|S(0)| \leq 0.1$ and $|T(0)| \leq 0.2$ in the tracking loop. Can clever weighting make both feasible?

Solution. No. Since $S + T = 1$, the triangle inequality would imply $1 \leq |S(0)| + |T(0)| \leq 0.3$. These constant DC requirements contradict the loop identity, regardless of controller order. Requesting small S at low frequency and small T at high frequency avoids this immediate contradiction, though it does not guarantee a feasible design for the given plant.

8.2 Convert an individual allowance into a joint one

Two output channels must each have gain at most 0.8. Does this certify their unit stack requirement? How can one certify it using only individual bounds?

Solution. No: the constant map $(0.8, 0.8)^T$ is a counterexample. Individual bounds b_1, b_2 suffice if $b_1^2 + b_2^2 \leq 1$; equal allocation requires $b_1 = b_2 = \frac{1}{\sqrt{2}}$. This is sufficient, not necessary for an arbitrary dynamic map: its peaks may not occur together. Compute the actual stacked singular value when conservatism matters.

8.3 What does scaling a weight actually do?

A fixed controller achieves $\|W_1(F - W_m)W_2\|_\infty = 0.6$. Replace W_1 by $2W_1$. Does the controller become unstable? Does it pass the unit test?

Solution. With the stable realizations and external topology just described, stability is unchanged. The new norm is exactly 1.2, so performance fails. The allowable physical discrepancy was halved. Scaling a requirement does not improve the physical loop; it changes the ruler used to judge it.

8.4 Run and audit

The adjacent `performance_examples.m` uses Octave's control package. Run it from any directory with `octave --no-gui --quiet <path>/performance_examples.m`. It recreates `assets/budgets.png`, checks the matrix products, phase-direction example, budget counterexamples and model-matching peak, and verifies the mixed stack by a nonnegative-polynomial certificate as well as a frequency grid. The plot is evidence of the shapes; endpoint and algebraic checks supply what a finite grid cannot prove. There is no controller synthesis or robust-uncertainty claim here: these are **nominal** specifications preparing the next chapters.

The source is Chapter 5, pp. 136–143, of [1]. The numerical examples, actuator-effort extension and realization counterexample are explanatory additions. No external research was needed.

Bibliography

[1] C. W. Scherer, "Theory of Robust Control." [Online]. Available: https://www.imng.uni-stuttgart.de/mst/files/RC.pdf